

EXAM 7 – SPRING 2011

8. (4 points)

Given the following cumulative paid loss information:

Accident Year	<u>Age of Development in Months</u>				
	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>
2006	\$223,000	\$303,000	\$354,000	\$376,000	\$381,000
2007	236,000	298,000	365,000	399,000	
2008	305,000	345,000	390,000		
2009	260,000	333,000			
2010	311,000				

	<u>12-Ult</u>	<u>24-Ult</u>	<u>36-Ult</u>	<u>48-Ult</u>	<u>60-Ult</u>
Estimators of loss development factors	1.599	1.280	1.092	1.013	1.000
Estimators of σ_j^2	2,410	716	173	42	0

- The estimation variance of the unpaid claims for accident year 2009 as of December 31, 2010 is \$157 million.

Use Mack's model, as described by England and Verrall, to calculate the prediction error as a percentage of the unpaid claims for accident year 2009 as of December 31, 2010.

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Sample Solutions
Exam 07, 2011, Question 8

Solution 1

a) Process variance equation for accident year(i) reserves¹

$$\text{Var}[R_i] \approx \hat{D}_{in}^2 \sum_{k=n-i+1}^{n-1} \frac{\hat{\sigma}_{k+1}^2}{\hat{\lambda}_{k+1}^2 \hat{D}_{ik}}$$

<ignoring the hats in the equations that follow, carrying decimal places from calculations>

$$D_{2009,5} = \text{Ultimate loss for AY 2009} = 333,000 \times 1.28 = 426,240$$

$$24-36 \text{ ldf} = 1.280/1.092 = 1.172$$

$$36-48 \text{ ldf} = 1.092/1.013 = 1.078$$

$$D_{2009,3} = D_{2009,2} \times (\text{ldf}_{24-36}) = 333,000 \times 1.172 = 390,330$$

$$D_{2009,4} = D_{2009,3} \times (\text{ldf}_{36-48}) = 390,330 \times 1.078 = 420,770$$

$$\text{Var}[R_{2009}] = 426,240^2 \times \left\{ \frac{173}{1.172^2 \times 333,000} + \frac{42}{1.078^2 \times 390,330} + \frac{0}{1.013^2 \times 420,770} \right\}$$

Therefore, process variance = 85,537,979

MSEP for AY2009 = process variance + prediction variance

$$\begin{aligned} &= 85,537,979 + 157,000,000 \\ &= 242,537,965 \end{aligned}$$

Prediction error = sqrt(MSEP) = 15,574

AY 2009 reserves = ultimate loss – current paid

$$\begin{aligned} &= 426,240 - 333,000 \\ &= 93,240 \end{aligned}$$

Prediction error as % of reserves for AY 2009 = 15,574 / 93,240 = 16.7%